Growth and development Overview

Third Stage

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Growth and Developments

LEARNING OBJECTIVES:

Mastery of content in this chapter will enable the student to:

1. Identify the importance of growth and development.
2. Describe and differentiate between growth and development.
3. Recognize and discuss the factors that influencing growth and development.
4. Describe the stages of development.

GROWTH AND DEVELOPMENT

Definitions:

- **Growth:**

It is the increase in physical size of the whole body or any of the parts of the body. Because of cell division and synthesis of proteins. It causes a quantitative change in the child's body.

Growth can be measured **accurately** in inches, centimeters, pounds or kilograms.

- **Development:**

  It refers to progressive increase in skills and capacity of functions.

  It causes a qualitative change in the child' functioning.

  Development can be measured but its measurements are not so accurate as growth as it is measured through observation.

Maturation:

Literally, it means ripen.

It is described as ageing or an increase in child's competence and adaptability.

It is usually used to describe qualitative changes in a structure, i.e., the structure begins to function or to function at a higher level depending on the child's heredity.

PRINCIPLES OF GROWTH AND DEVELOPMENT:

1. Growth and development are continuous processes.
2. Each child has his own rate of growth and development.
3. Both growth and development are synchronous. Different body parts and developmental areas develop at different rates. Each body part has its own time to grow.
4. All areas of growth and development are interrelated (physical, psychosocial, cognitive and motor) to meet the child's basic physiological needs.

5. Children's behavior is directed towards competency.
6. Growth and development progress from midline towards periphery.
7. Growth and development occur in an orderly sequence.
8. The direction of development is Cephalocaudal, i.e. the infant controls his head before he can sit, and crawls before he can walk.
9. General to specific: at the beginning all faces and sounds appear same to him/her, then stars to distinguish mother's face and sound.
10. Development is relative

Factors affecting growth and development:

1. Hereditary factors: These are responsible for the characteristics of :
    - certain anomalies,
    - individual pattern of growth,
    - some diseases
    - certain types of short stature.
2. Congenital defects: These may be due to heredity or abnormal environment during embryonic life.
3. Endocrine factors: These can produce normal variations and cause growth variation.

4. *Environmental factors: Environmental factors such as
   - the mother's nutritional status,
   - exposure to infectious diseases and pathological conditions
   - the socioeconomic conditions of the family affect the health of the embryo, infant and child.
5. *_Nutritional factors_: Growth of the baby depends on adequate consumption and proper utilization of suitable types of food. Specific deficiencies cause alterations in growth and development.

6. ***Activities:*** **Normal growth and development depend on normal activities. A bedridden child would not grow as much as an active one.**
7. ***Illness and injuries:*** **During birth, may cause a permanent damage to vital organs.**

Types of Growth:

1. Physical growth such as, weight, height, head and chest circumferences.
2. Physiological growth such as, vital signs, and senses; (touch, hearing, smell, vision, and taste).

Types of Development:

1. Cognitive development (intellectual development e.g. reasoning, mental thinking…etc)
2. Emotional development e.g. loves, fear …etc.
3. Social development e.g. self concept, friendship, and development of senses such as sense of trust…etc.

STAGES OF GROWTH AND DEVELOPMENT

1. **Prenatal period : From conception to birth:**

I. Germinal period:1st two weeks of gestation (Zygote)

II. Embryonic period: 2 – 8 weeks (Embryo)

III. Fetal period: 8weeks – birth (Fetus)

2. **Infancy: Birth to 12 months**
   I. Neonate : Birth to 1 month
   II. Infancy: 1 month to 12 months
3. **Early Childhood: 1to 6 years**
   I. Toddler: 1-3 years
   II. Preschool: 3-6 years

4. **Middle Childhood: 6 to 11 0r 12 years**
   - School age:
     6 to 12 years
5. **Later Childhood: 11 to 19**
   I. Prepubertal: 10 -13 years
   II. Adolescence: 13 years to approximately 18 or 19 years

6. Adulthood

I. Young adulthood: from 19- 25 years
II. Adult: from 26 – 40 years
III. Maturity: from 41 years - and above

7. Old age

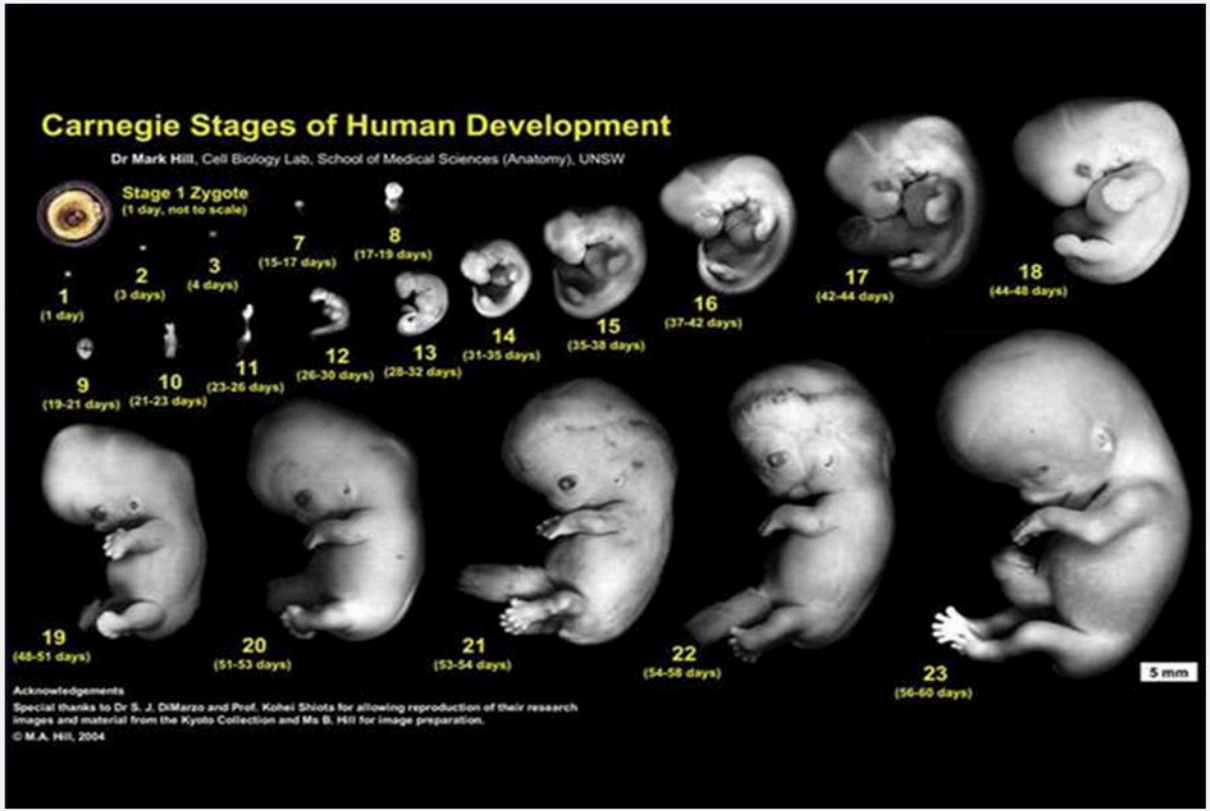
Table 3.1 Human Growth Sequence

Period	Duration	Descriptive Name
Prenatal Period	From conception to birth	
Germinal period	First 2 weeks after conception	Zygote
Embryonic period	2–8 weeks after conception	Embryo
Fetal period	From 8 weeks after conception to birth	Fetus
Neonatal Period	From birth to a few weeks after birth	Neonate
Infancy	From a few weeks after birth until child is walking securely; some children walk securely at less than a year, while others may not be able to until age 17–18 months	Infant
Early Childhood	From about 15–18 months until about 2–2½ years	Toddler
	From age 2–3 to about age 6	Preschool child
Middle Childhood	From about age 6 to age 12	School-age child
Pubescence	Period of about 2 years before puberty	
Puberty	Point of development at which biological changes of pubescence reach a climax marked by sexual maturity	
Adolescence	From the beginning of pubescence until full social maturity is reached (difficult to fix duration of this period)	Adolescent
Adulthood	From adolescence to death; sometimes subdivided into other periods as shown at left	Adult
Young adulthood (19–25)		
Adulthood (26–40)		
Maturity (41 plus)		
Senescence	No defined limit that would apply to all people; extremely variable; characterized by marked physiological and psychological deterioration	Adult (senile), "old age"

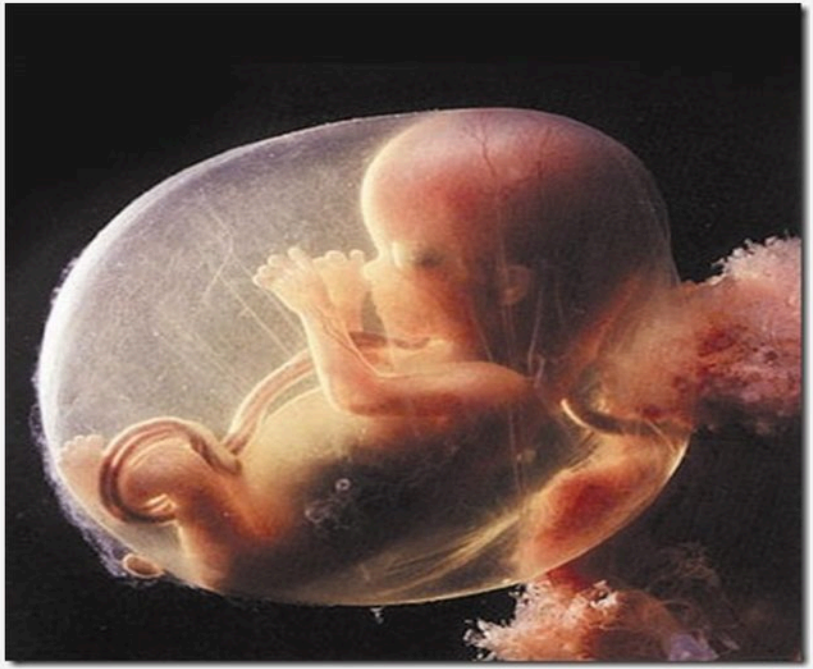
**Note:* There is no exact beginning or ending point for various growth periods. The ages are approximate, and each period may be thought of as blending into the next. (Table courtesy of Tom Bond.)

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EMBRYO


Carnegie Stages of Human Development
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Stage 1 Zygote
(1 day, not to scale)
1
(1 day)
2
(3 days)
3
(4 days)
7
(15-17 days)
8
(17-19 days)
9
(19-21 days)
10
(21-23 days)
11
(23-26 days)
12
(26-30 days)
13
(28-32 days)
14
(31-35 days)
15
(35-38 days)
16
(37-42 days)
17
(42-44 days)
18
(44-48 days)
19
(48-51 days)
20
(51-53 days)
21
(53-54 days)
22
(54-56 days)
23
(56-60 days)
5 mm
Acknowledgements
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Fetus (12 weeks)



Newborn infant



Infant



Toddler



Preschool



School child



Preadolescent (Prepubertal)



Adolescent



Young adult



Mature (Older adult)



Old aged





Thank you

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